

# Capacity Triage: How Normalization Choice Shapes Per-Class Accuracy in Width-Varying CNNs

Anonymous Author(s)

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## Abstract

Batch Normalization (BN) reliably improves aggregate accuracy, yet its effect on the *distribution* of accuracy across classes is poorly understood. We sweep network width to vary the ratio of representation dimension to class count ( $d/K$ ), using a matched-accuracy Gini analysis as an auxiliary tool to control for overall accuracy differences when comparing per-class allocation under different normalization strategies. We find a dual-layer structure: BN consistently outperforms Group Normalization across widths and architectures ( $\sim 2.3$ – $4.4$  pp on ScaleCNN/Tiny-ImageNet;  $\sim 4.1$ – $6.6$  pp on ResNet), but its effect on per-class allocation is reproducible only above a  $d/K$  threshold. Ten-seed validation shows that the standardized effect size (Cohen’s  $d$ ) is consistently large ( $d \geq 1.88$ ) across all widths with 10-seed validation ( $c \in \{8, 10, 12, 16, 32\}$ ), with effect directions replicating on Tiny-ImageNet ( $K=200$ ) and strong statistical significance on ResNet-small ( $d=4.80$ ,  $p < 10^{-8}$  at  $c=32$ ). The matched-accuracy Gini delta transitions from stochastic (seed-irreproducible) to deterministic (seed-stable) near  $d/K \approx 0.7$ , suggesting that the  $d/K$  ratio serves as an additional predictor of when normalization choice will deterministically shape class-level outcomes.

## 1 Introduction

Normalization layers—Batch Normalization (BN) [Ioffe and Szegedy, 2015], Layer Normalization (LN) [Ba et al., 2016], Group Normalization (GN) [Wu and He, 2018], Instance Normalization (IN) [Ulyanov et al., 2016]—are standard components of modern deep networks. The choice among them is typically guided by training considerations: BN for large-batch CNNs, LN for Transformers, GN when batch size is constrained. A question that has received far less attention is whether this choice affects the *distribution* of accuracy across classes. Two models with identical overall accuracy can exhibit markedly different per-class profiles—some classes improving at the expense of others—a phenomenon we term **capacity triage**. In sub-million parameter networks, where representational capacity is limited, such triage effects may be particularly pronounced.

Prior work has studied how normalization affects the optimization landscape [Santurkar et al., 2018], training speed [Ioffe and Szegedy, 2015], and aggregate accuracy, but has not systematically examined how normalization shapes the per-class accuracy distribution. Separately, Hooker et al. [2020] demonstrated that model compression can disproportionately degrade underrepresented classes, establishing that architectural choices can have class-specific consequences. Yet no prior study isolates the effect of normalization type on per-class allocation, and practitioners lack guidance on when this effect is large enough to matter.

We address this gap with the following contributions:

- **Large, reproducible effect sizes.** Extending the per-class analysis framework of Alnemari et al. [2026] to the normalization dimension, we show that BN’s effect on per-class allocation is consistently large (Cohen’s  $d \geq 1.88$ ) across all widths with 10-seed validation

( $c \in \{8, 10, 12, 16, 32\}$ ), with variance decreasing faster than the gap (Section 4.3). A  $d/K$  reproducibility threshold near  $\sim 0.7$  marks a transition from stochastic to deterministic triage.

- **Partial replication on Tiny-ImageNet.** The direction of all effects replicates on Tiny-ImageNet ( $K=200$ ), with a BN-vs-GN comparison at two widths above the crossover (Section 4.5).
- **Cross-architecture validation on ResNet-small.** On ResNet-small (with skip connections), the BN-GN accuracy gap is 4.11–6.64 pp across all three widths—larger than the ScaleCNN gap at matched  $d/K$ , with strong statistical significance at  $c=32$  (Cohen’s  $d=4.80$ ,  $p < 10^{-8}$ , 10 seeds; Section 4.6).

In brief, BN’s effect on per-class allocation is consistently large (Cohen’s  $d \geq 1.88$ ) and consistent across both tested architectures (ScaleCNN and ResNet-small), with the BN-GN gap positive across all tested widths and both architectures. A  $d/K$  reproducibility threshold near  $\sim 0.7$  marks the transition from stochastic to deterministic triage (Section 4.3).

## 2 Related Work

**Normalization methods.** BN [Ioffe and Szegedy, 2015], LN [Ba et al., 2016], GN [Wu and He, 2018], and IN [Ulyanov et al., 2016] differ in the dimensions over which they compute statistics. Their *training dynamics* are well-studied—loss landscape smoothness [Santurkar et al., 2018], gradient flow [Bjorck et al., 2018], representation clustering [Potgieter et al., 2025]—but their impact on per-class accuracy distribution has received little attention. Bao et al. [2025] show that standard BN statistics are biased in long-tailed settings and propose re-normalization; we study balanced datasets where this bias does not apply. Recent work has shown that normalization can be removed entirely: Zhu et al. [2025] replace normalization layers with a learned tanh activation (DyT), matching normalized performance without batch or layer statistics. Our work differs from this line in that we study how normalization *redistributes* accuracy across classes, rather than whether it can be replaced.

**Scaling and per-class error structure.** Concurrent work by Alnemari et al. [2026] uses the same tools—a plain CNN they also call ScaleCNN, CIFAR-100, and the per-class accuracy Gini coefficient—to study how width scaling affects per-class error patterns. Their focus is the relationship between model scale and per-class mistakes; ours is the effect of *normalization type* on per-class allocation at fixed scale. The research questions are complementary but distinct: they vary width with a single normalization; we vary normalization at each width. Model-scaling effects in CNNs are further studied by Tan and Le [2019] (compound scaling) and Zagoruyko and Komodakis [2016] (width scaling with residual connections [He et al., 2016]).

**Neural collapse and feature geometry.** Papayan et al. [2020] identified neural collapse—the convergence of last-layer features to a simplex equiangular tight frame—during terminal training. Zhu et al. [2021] provided a geometric analysis with unconstrained features, and Jiang et al. [2024] extended the theory to the regime  $K \gg d$ , proving generalized collapse that maximizes the minimum one-vs-rest margin. This regime is complementary to the  $d/K > 0.8$  threshold we identify: neural collapse describes feature geometry at convergence, while our crossover concerns *reproducibility* of per-class allocation across seeds.

**Fairness in per-class performance.** Hooker et al. [2020] showed that compression disproportionately degrades underrepresented classes. Subsequent work addressed class imbalance through data resampling [Buda et al., 2018], augmentation [Shorten and Khoshgoftaar, 2019], loss reweighting [Cui et al., 2019, Cao et al., 2019], logit adjustment [Menon et al., 2021], and distributionally robust optimization [Sagawa et al., 2020]. Unlike these studies, which focus on data-side or post-hoc interventions for imbalanced settings, we isolate a purely architectural factor—normalization type—and show it shapes per-class allocation even on balanced datasets.

**Threshold phenomena.** Emergent abilities [Wei et al., 2022] and double descent [Nakkiran et al., 2020] show that qualitative behavior can change abruptly with scale. Our crossover concerns *reproducibility* rather than *emergence*: BN affects per-class allocation at all widths, but this effect is seed-stable only above the  $d/K$  threshold.

## 3 Experimental Setup

### 3.1 Architecture

We use **ScaleCNN**, a four-block convolutional network with channel progression  $[c, 2c, 4c, 8c]$ , where  $c$  is the base width. (ScaleCNN refers to our custom architecture; concurrent work by Alnemari et al. [2026] independently uses a similarly named architecture for studying scaling laws.) Each block consists of two  $3 \times 3$  convolution layers followed by normalization (or none) and ReLU activation. Global average pooling after the final block yields a feature vector of dimension  $d = 8c$ , which feeds a linear classifier. The architecture contains no skip connections and no attention—these omissions are deliberate, isolating the effect of normalization from residual or attention-mediated interactions.

On CIFAR-100 ( $K=100$ ), we sweep  $c \in \{4, 8, 10, 12, 16, 24, 32, 48\}$ , producing  $d/K$  ratios from 0.32 to 3.84. On Tiny-ImageNet ( $K=200$ ), we use  $c \in \{32, 48\}$ , yielding  $d/K \in \{1.28, 1.92\}$ . For cross-architecture validation, we train a ResNet-small variant (four residual blocks,  $c \in \{8, 16, 32\}$ ) on CIFAR-100 with BN, GN, and no normalization, using the same training protocol (three seeds each, 27 runs; augmented to 10 seeds at  $c=32$  for BN and GN).

### 3.2 Training Configuration

All models are trained with SGD (momentum 0.9, weight decay  $5 \times 10^{-4}$ ) for 200 epochs with cosine annealing and batch size 128. The initial learning rate is 0.1 for BN and 0.05 for GN, LN, IN, and no normalization; we selected the non-BN rate via a learning-rate sweep over  $\{0.01, 0.03, 0.05\}$ : GN performed best at 0.05, while BN achieved comparable accuracy at both 0.1 and 0.05 (e.g., 66.96% vs. 66.42% at  $c=32$ ), indicating that non-BN variants require a lower learning rate.

Data augmentation consists of random cropping with 4-pixel padding and random horizontal flipping. CIFAR-100 uses  $32 \times 32$  images with 100 classes (50k training / 10k test); Tiny-ImageNet uses  $32 \times 32$  images (resized from native  $64 \times 64$  for architectural compatibility with ScaleCNN) with 200 classes (100k training / 10k validation). Each configuration is run with three random seeds (42, 123, 456). For statistical reliability at key widths ( $c \in \{8, 10, 12, 16, 32\}$ ), we augment to ten seeds per condition (seeds 42–51); these *10-seed* results are used for all accuracy and Gini claims in the text (Welch’s  $t$ -test, Cohen’s  $d$ ), while tables based on the original 3-seed width curve are annotated accordingly. The GN group count is  $g=8$  for  $c \geq 16$  and  $g=2$  for  $c=8$  in all 10-seed experiments; the 3-seed width curve used  $g=4$  at some intermediate widths (see individual table captions for details). We ran all experiments on NVIDIA RTX 5090 GPUs (32 GB); a single ScaleCNN run at  $c=32$  takes  $\sim 8$  minutes, and the full suite ( $\sim 280$  runs) required approximately 28 GPU-hours. Code will be made available upon acceptance.

### 3.3 Metrics and Design

For each class  $i$ , per-class accuracy is  $a_i = n_i^{\text{correct}} / n_i^{\text{total}}$ . We measure concentration with the Gini coefficient:

$$G = \frac{\sum_{i=1}^K \sum_{j=1}^K |a_i - a_j|}{2K \sum_{i=1}^K a_i}, \quad (1)$$

where  $G=0$  is perfectly uniform and  $G \rightarrow 1$  is fully concentrated. To compare distributions at equal accuracy, we interpolate each variant’s Gini-vs-accuracy trajectory and compute  $\Delta G = G_{\text{BN}}(a_{\text{match}}) - G_{\text{baseline}}(a_{\text{match}})$  (details in Section 4.2). Cross-seed reproducibility is  $\text{SNR} = |\Delta G| / \sigma_{\Delta G}$ ; a width is *reliable* if  $\text{SNR} > 1.0$ .

We evaluate five normalization conditions: BN, GN (groups = 8 for  $c \geq 16$ ; groups = 2 for  $c=8$ ), LN, IN, and None. The primary width curve compares BN against GN; for widths where GN’s group parameter does not divide channels evenly ( $c=4, 10, 12$ ), we use the nearest feasible group count.

## 4 Results

### 4.1 BN’s Accuracy Advantage on ScaleCNN is Width-Persistent

BatchNorm’s accuracy advantage over Group Normalization persists across all tested network widths on ScaleCNN. Across the full width curve (eight widths from  $c=4$  to  $c=48$ ; baseline: GN), BN’s final accuracy exceeds GN at every tested width, with no monotonic dependence on  $d/K$ .

Ten-seed validation confirms consistent BN advantage: at  $c=32$  ( $d/K=2.56$ ), BN achieves 66.92% versus GN’s 64.07% (+2.86 pp); at  $c=16$  ( $d/K=1.28$ ), 60.76% versus 56.35% (+4.40 pp); at  $c=8$  ( $d/K=0.64$ ), 49.76% versus 47.03% (+2.73 pp). The gap varies from 2.73 to 4.40 pp across widths but shows no monotonic trend with  $d/K$ ; the key finding is that BN’s advantage persists at every tested width rather than being numerically constant. On Tiny-ImageNet ( $K=200$ ), the BN-vs-GN gap is likewise width-stable: +2.26 pp at  $c=32$  ( $d/K=1.28$ ) and +2.55 pp at  $c=48$  ( $d/K=1.92$ ) (Section 4.5).

The BN-GN gap is positive across all tested widths and both architectures; its magnitude varies (larger on ResNet-small, Section 4.6) but the direction is consistent across both tested architectures (ScaleCNN and ResNet-small). This is important context for the per-class results that follow. To ask whether BN distributes accuracy differently across classes, we first introduce a matched-accuracy methodology that controls for the accuracy gap itself (Section 4.2), then use it to identify a  $d/K$  reproducibility threshold (Section 4.3).

### 4.2 Matched-Accuracy Gini Analysis

Raw Gini comparisons between BN and GN are confounded by overall accuracy differences: higher accuracy mechanically compresses the per-class distribution toward the mean, reducing Gini independently of any change in allocation strategy. To isolate normalization’s effect on *how* accuracy is distributed across classes, we propose a **matched-accuracy Gini** comparison.

For each width  $c$  and each random seed, we train both BN and GN variants for 200 epochs, recording per-class accuracy and overall accuracy at every epoch. We define the matched accuracy level as the final accuracy of the lower-performing variant (typically GN). We then interpolate the BN variant’s Gini-vs-accuracy trajectory to obtain its Gini at this matched accuracy level, which corresponds to an intermediate training epoch rather than convergence. The matched-accuracy Gini delta is

$$\Delta G = G_{\text{BN}}(a_{\text{match}}) - G_{\text{GN}}(a_{\text{match}}), \quad (2)$$

where a positive  $\Delta G$  indicates that BN concentrates per-class accuracy more aggressively than the baseline at the same overall accuracy.

At  $c=32$  ( $d/K=2.56$ ), three seeds yield  $\Delta G$  values of 0.0117, 0.0102, and 0.0088, with mean 0.0102, standard deviation 0.0015,  $\text{SNR} = 7.05$ , and  $p=0.006$  (two-sided  $t$ -test). The sign is positive across all seeds, indicating that BN reproducibly produces a more concentrated per-class distribution at matched accuracy.

**Sensitivity caveat.** Because the matched-accuracy comparison evaluates BN at an intermediate epoch, the precise  $\Delta G$  magnitude is epoch-dependent: as the matched epoch varies from approximately 147 to 168 across seeds and hyperparameter choices,  $\Delta G$  ranges from 0.0088 to 0.0117. Critically, the *sign* of  $\Delta G$  is stable—BN always shows higher Gini at matched accuracy—but the magnitude should be interpreted with this sensitivity in mind. We therefore treat the matched-accuracy analysis as an *auxiliary* method that confirms the direction of BN’s effect. The primary evidence for a  $d/K$ -dependent threshold comes from the width curve (Section 4.3), which tests *reproducibility* across seeds rather than relying on a single matched-accuracy estimate.

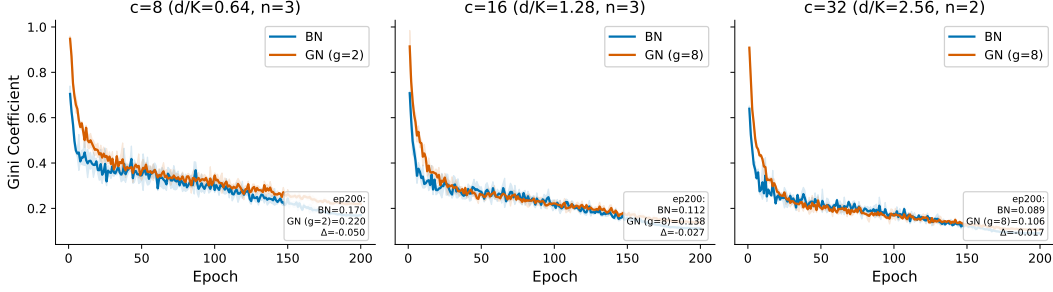


Figure 1: Gini coefficient trajectory during training for BN (blue) vs. GN (orange) across three widths:  $c=8$  ( $g=2$ ,  $n=3$ ),  $c=16$  ( $g=8$ ,  $n=3$ ),  $c=32$  ( $g=8$ ,  $n=2$ ). BN converges to lower Gini at epoch 200 across all widths, indicating more uniform per-class accuracy distributions. Shaded regions show  $\pm 1$  standard deviation across seeds.

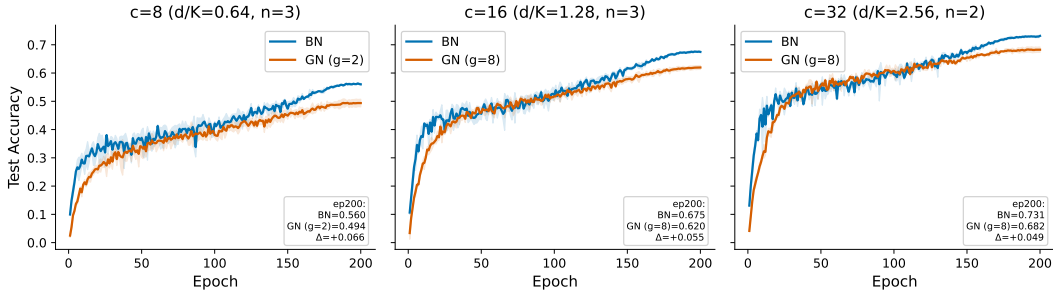


Figure 2: Accuracy trajectory during training for BN vs. GN, with the same layout as Figure 1. BN achieves higher final accuracy at all three widths. The matched-accuracy analysis (Section 4.2) uses the intersection of BN’s trajectory with GN’s final accuracy to define the matched epoch.

### 4.3 The $d/K$ Reproducibility Threshold

BN’s effect on per-class allocation transitions from stochastic to deterministic as  $d/K$  increases past a crossover interval. We establish this by varying the base channel count  $c$  of ScaleCNN from 4 to 48, sweeping  $d/K$  from 0.32 to 3.84 ( $d=8c$ ,  $K=100$ ). For each width, we train BN and GN variants with three random seeds and compute the matched-accuracy Gini delta  $\Delta G$  per seed. Table 1 and Figure 3 report the mean delta, standard deviation, SNR ( $= |\Delta G|/\sigma$ ), mean matched epoch, and reproducibility verdict.

**Crossover interval.** Below  $d/K \approx 0.8$ , the matched-accuracy Gini delta is positive but noisy: at  $c=4$  ( $d/K=0.32$ ) and  $c=8$  ( $d/K=0.64$ ), SNR is 0.31 and 0.50 respectively, indicating that  $\Delta G$  is indistinguishable from seed-level variation. At  $c=10$  ( $d/K=0.80$ ), SNR jumps to 4.98, and all widths with  $d/K \geq 0.80$  show  $\text{SNR} > 1.4$ . We report the crossover as the interval  $d/K \in [0.64, 0.96]$ , bounded by the last unreliable width ( $c=8$ ) and the first width at which a conservative reader might still question reliability ( $c=12$ ,  $\text{SNR} = 1.42$ ). A nonparametric bootstrap (10,000 resamples; Figure 8, Appendix A) confirms that the crossover is sharply localized: 77.8% of bootstrap samples place the transition within  $d/K \in [0.64, 0.80]$ .

**Anti-artifact evidence.** A potential confound is that the matched epoch could itself drive the SNR transition. Table 1 rules this out: at  $c=8$  (unreliable,  $\text{SNR} = 0.50$ ) the mean matched epoch is 170.7, while at  $c=10$  (reliable,  $\text{SNR} = 4.98$ ) it is 171.3—a difference of 0.6 epochs, yet SNR differs by  $10\times$ . The crossover is associated with network width, not optimization stage.

Table 1: Width curve: matched-accuracy Gini delta (BN vs. GN) on CIFAR-100 ( $K=100$ ), 3 seeds per width. GN uses  $g=2$  at  $c=8$ ,  $g=4$  at  $c \in \{10, 12\}$ , and  $g=8$  at  $c \geq 16$ . Verdict: *reliable* if  $\text{SNR} > 1.0$ , *unreliable* otherwise.  $c=12$  is borderline ( $\text{SNR} = 1.42$ ).

| $c$ | $d$ | $d/K$ | $\overline{\Delta G}$ | $\sigma$ | SNR  | Matched epoch |
|-----|-----|-------|-----------------------|----------|------|---------------|
| 4   | 32  | 0.32  | 0.0030                | 0.0097   | 0.31 | $\sim 168$    |
| 8   | 64  | 0.64  | 0.0041                | 0.0082   | 0.50 | 170.7         |
| 10  | 80  | 0.80  | 0.0027                | 0.0005   | 4.98 | 171.3         |
| 12  | 96  | 0.96  | 0.0047                | 0.0033   | 1.42 | $\sim 165$    |
| 16  | 128 | 1.28  | 0.0053                | 0.0006   | 9.34 | $\sim 163$    |
| 24  | 192 | 1.92  | 0.0127                | 0.0022   | 5.88 | $\sim 155$    |
| 32  | 256 | 2.56  | 0.0102                | 0.0015   | 7.05 | $\sim 165$    |
| 48  | 384 | 3.84  | 0.0132                | 0.0039   | 3.35 | $\sim 142$    |

Table 2: Cohen’s  $d$  for the BN–GN Gini gap across all 10-seed widths (ScaleCNN on CIFAR-100) and ResNet-small at  $c=32$ . All effects are large ( $d \geq 1.88$ ) with  $p < 10^{-3}$ .

| Arch     | $c$ | $d/K$ | Gini gap | Cohen’s $d$ | $p$ -value  | Seeds |
|----------|-----|-------|----------|-------------|-------------|-------|
| ScaleCNN | 8   | 0.64  | 0.0177   | 1.88        | $< 10^{-3}$ | 10    |
| ScaleCNN | 10  | 0.80  | 0.0164   | 3.02        | $< 10^{-4}$ | 10    |
| ScaleCNN | 12  | 0.96  | 0.0144   | 3.37        | $< 10^{-5}$ | 10    |
| ScaleCNN | 16  | 1.28  | —        | 1.99        | $< 10^{-3}$ | 10    |
| ScaleCNN | 32  | 2.56  | 0.0091   | 3.55        | $< 10^{-6}$ | 10    |
| ResNet   | 32  | 2.56  | 0.0154   | 4.80        | $< 10^{-8}$ | 10    |

**Stochastic-to-deterministic interpretation.** The crossover marks a transition in *reproducibility*, not existence. At low  $d/K$ , BN still improves overall accuracy, but per-class triage patterns vary across seeds. Above  $d/K \approx 0.8$ ,  $\Delta G$  is consistently positive with low variance. Low- $d/K$  triage is *stochastic* (seed-chaotic); high- $d/K$  triage is *deterministic* (seed-reproducible).

**Effect size is consistently large.** Ten-seed validation at  $c \in \{8, 10, 12, 16, 32\}$  reveals that the *absolute* Gini gap (BN minus GN) narrows with width, yet the *standardized effect size* (Cohen’s  $d$ ) is consistently large ( $d \geq 1.88$ ) across all five widths (Table 2):  $d = 1.88$  at  $c=8$  ( $d/K=0.64$ ), 3.02 at  $c=10$  ( $d/K=0.80$ ), 3.37 at  $c=12$  ( $d/K=0.96$ ), 1.99 at  $c=16$  ( $d/K=1.28$ ), and 3.55 at  $c=32$  ( $d/K=2.56$ ) (all  $p < 10^{-3}$ , Welch’s  $t$ -test). Variance decreases faster than the gap shrinks, so the signal-to-noise ratio improves with  $d/K$ . This pattern is consistent with the  $d/K$  framework: as capacity surplus grows, per-seed variance decreases and the normalization-dependent triage pattern becomes increasingly detectable, even as its absolute magnitude diminishes.

**Dual-layer finding.** Combining Sections 4.1 and 4.3 yields the dual-layer structure of our results. BN improves overall accuracy by a consistent  $\sim 2.3$ – $4.4$  pp (depending on width and dataset) at all tested widths—on ScaleCNN, this effect is width-independent; on ResNet, the gap is larger and width-dependent (4.11–6.64 pp; see Section 4.6). BN’s effect on the per-class accuracy *distribution*, however, is width-dependent: it is reproducible only when  $d/K$  exceeds the crossover interval. The  $d/K$  ratio thus serves as an empirical predictor of when normalization choice will have a *deterministic* effect on class-level outcomes, distinct from its *always-present* effect on aggregate accuracy.

**Baseline selection.** We use GN as baseline throughout because it is the most widely used practical alternative to BN, isolating BN-specific from normalization-general effects. Section 4.4 reports a five-way comparison.

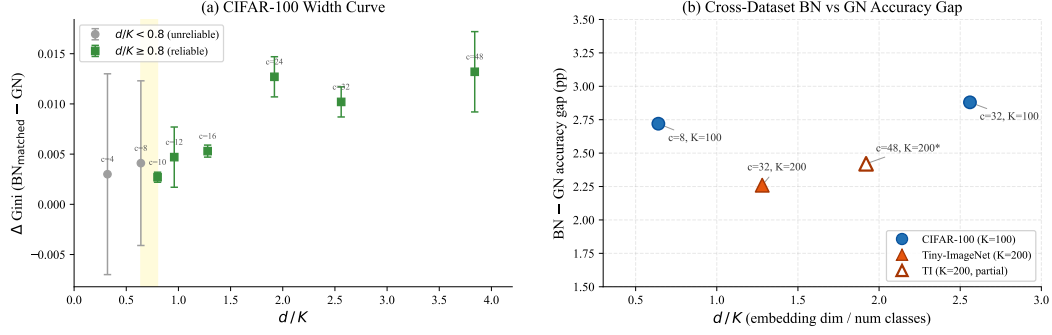


Figure 3: (a) Matched-accuracy Gini delta ( $\Delta G$ , BN minus GN at matched accuracy) across eight ScaleCNN widths on CIFAR-100. Green squares indicate reliable widths ( $\text{SNR} > 1.0$ ); gray circles are unreliable. The yellow band marks the crossover interval  $d/K \in [0.64, 0.96]$ . Error bars show  $\pm 1$  standard deviation across three seeds. (b) BN–GN accuracy gap (pp) as a function of  $d/K$  across CIFAR-100 ( $K=100$ , blue circles) and Tiny-ImageNet ( $K=200$ , orange triangles). The gap is positive at all widths with no systematic trend, ranging from  $\sim 2.4$ – $3.5$  pp across most widths (3-seed width curve; 10-seed validation at  $c \in \{8, 16, 32\}$  yields 2.73–4.40 pp, see Section 4.1).

Table 3: Normalization spectrum on CIFAR-100. Accuracy and raw Gini coefficient (3-seed means) at two widths; GN  $c=32$  accuracy uses 10-seed canonical (64.07%, rounded to 0.641). Lower Gini indicates more uniform per-class accuracy distribution. GN uses sweep-optimal group count ( $g=2$  at  $c=8$ ,  $g=8$  at  $c=32$ ).

|      | $c=8$ ( $d/K=0.64$ ) |              | $c=32$ ( $d/K=2.56$ ) |              |
|------|----------------------|--------------|-----------------------|--------------|
| Norm | Acc                  | Gini         | Acc                   | Gini         |
| BN   | <b>0.497</b>         | <b>0.197</b> | <b>0.670</b>          | <b>0.115</b> |
| LN   | 0.479                | 0.215        | 0.651                 | 0.124        |
| GN   | 0.469                | 0.215        | 0.641                 | 0.124        |
| None | 0.475                | 0.206        | 0.592                 | 0.146        |
| IN   | 0.377                | 0.292        | 0.578                 | 0.161        |

#### 4.4 Normalization Spectrum

BN occupies the Pareto-optimal corner (highest accuracy, lowest Gini) across five normalization strategies—Batch Normalization (BN), Layer Normalization (LN), Group Normalization (GN), Instance Normalization (IN), and no normalization (None)—at two widths that bracket the  $d/K$  threshold identified in Section 4.3. Table 3 reports accuracy and raw Gini coefficient (3-seed means) at  $c=8$  ( $d/K=0.64$ ) and  $c=32$  ( $d/K=2.56$ ).

At high capacity ( $c=32$ ,  $d/K=2.56$ ), the Gini ranking is  $\text{BN} < \text{LN} \approx \text{GN} < \text{None} < \text{IN}$ , with a clear gap between normalized variants (BN/LN/GN, all  $\text{Gini} \leq 0.124$ ) and the unnormalized baseline ( $\text{Gini} = 0.146$ ). At low capacity ( $c=8$ ,  $d/K=0.64$ ), the ranking shifts to  $\text{BN} < \text{None} < \text{LN} \approx \text{GN} < \text{IN}$ : the norm-free variant now achieves lower Gini than LN and GN, and the spread among BN, LN, GN, and None narrows (Gini range 0.197–0.215 vs. 0.115–0.146 at  $c=32$ ). This narrowing is consistent with the stochastic regime identified in Section 4.3: below the  $d/K$  threshold, the normalization-induced structure in per-class allocation becomes less distinguishable from seed variance.

Two patterns hold at both widths: BN occupies the Pareto-optimal corner (highest accuracy, lowest Gini), while IN yields the worst on both axes—consistent with its design for style transfer [Ulyanov et al., 2016] rather than classification. LN and GN produce nearly identical Gini (within 0.001), suggesting channel-grouping granularity matters less than normalization scope. A per-class

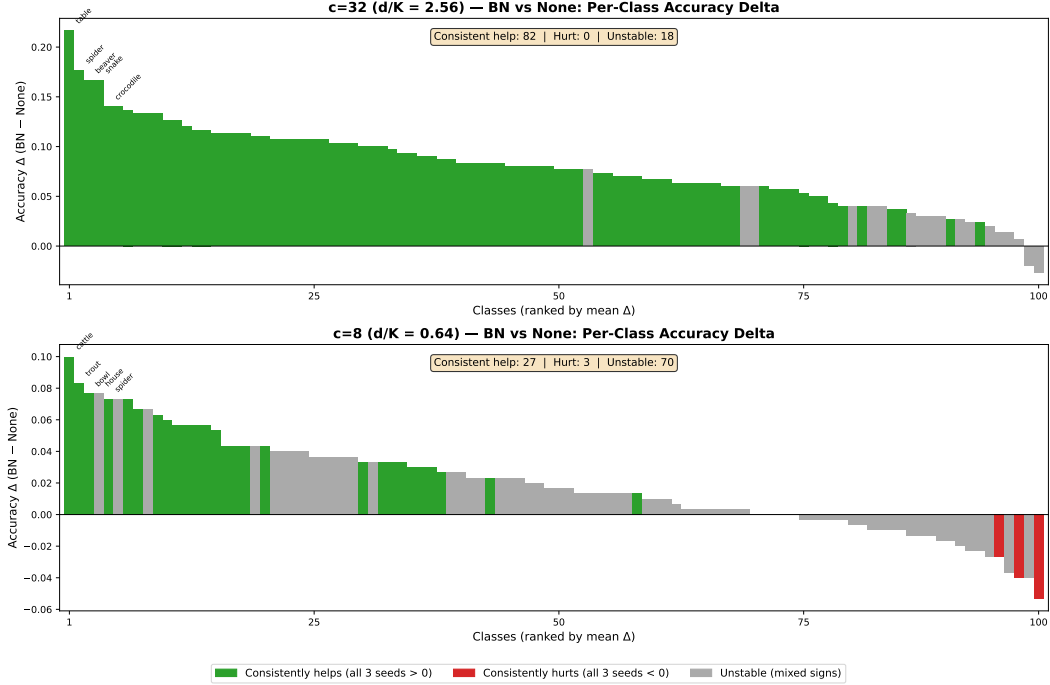


Figure 4: Per-class accuracy delta (BN minus None) for 100 CIFAR-100 classes. Top:  $c=32$ ; bottom:  $c=8$ . Green/red/gray = consistently helped/hurt/unstable across 3 seeds. At  $c=32$ , 82 classes helped, none hurt; at  $c=8$ , 27 helped, 3 hurt, 70 unstable—consistent with the stochastic regime below the crossover.

Table 4: Early-stopping analysis: BN vs. GN accuracy at four training checkpoints (ScaleCNN  $c=32$ , CIFAR-100, 10 seeds each). GN leads significantly at epochs 50 and 100; BN overtakes at epoch 150 and achieves a large advantage by convergence.  $p$ -values from Welch’s  $t$ -test.

| Epoch | BN Acc (%)       | GN Acc (%)       | Gap (pp) | $p$ -value  |
|-------|------------------|------------------|----------|-------------|
| 50    | $49.37 \pm 2.24$ | $52.83 \pm 1.05$ | $-3.46$  | $< 0.001$   |
| 100   | $54.15 \pm 1.22$ | $56.72 \pm 0.65$ | $-2.57$  | $< 10^{-4}$ |
| 150   | $61.71 \pm 0.27$ | $61.04 \pm 0.40$ | $+0.67$  | $< 0.001$   |
| 200   | $66.92 \pm 0.29$ | $64.07 \pm 0.32$ | $+2.85$  | $< 10^{-6}$ |

breakdown (Figure 4) confirms that at  $c=32$ , BN helps 82 of 100 classes with no class consistently hurt, whereas at  $c=8$  the pattern is dominated by seed variance.

These associations are descriptive; the accuracy–Gini confound (Section 4.2) applies here, and a matched-accuracy analysis across all five types is left to future work.

**Convergence crossover.** Under each norm’s optimized hyperparameters (BN  $\text{lr}=0.1$ , GN  $\text{lr}=0.05$ ), GN achieves higher intermediate accuracy (epochs  $\sim 15$ – $141$ ) while BN attains a significantly higher final accuracy ( $+2.85$  pp, 10/10 win rate; Figure 6). BN’s advantage is thus a *convergence-phase* property, emerging only in the last  $\sim 60$  epochs. Table 4 quantifies this: at epochs 50 and 100, GN leads by 3.46 and 2.57 pp respectively ( $p < 0.001$ ); BN overtakes at epoch 150 ( $+0.67$  pp,  $p < 0.001$ ) and widens the gap to  $+2.85$  pp at convergence. The early GN lead is partially attributable to its lower learning rate providing smoother early optimization; our LR confound control (Appendix D: BN at  $\text{lr}=0.05$  yields only  $-0.09\%$  final accuracy relative to  $\text{lr}=0.1$ ) confirms this does not affect the converged result.



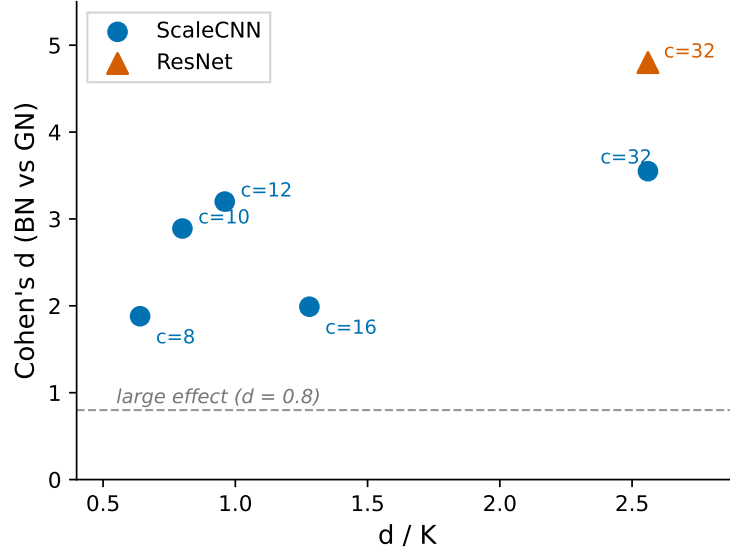


Figure 5: Standardized effect size (Cohen’s  $d$ ) for the BN–GN Gini gap as a function of  $d/K$ . All widths show  $d \geq 1.88$  (large effect), with the ResNet  $c=32$  point ( $d=4.80$ ) exceeding the ScaleCNN benchmark at matched  $d/K$ .

Table 5: BN vs. GN on Tiny-ImageNet ( $K=200$ ), 3 seeds per cell. Gap is BN accuracy minus GN accuracy.  $\Delta G_{\text{raw}} = G_{\text{BN}} - G_{\text{GN}}$ ; negative values indicate BN produces a more uniform distribution (lower Gini).

| $c$ | $d/K$ | BN acc | GN acc | Gap (pp) | BN Gini | GN Gini | $\Delta G_{\text{raw}}$ | Seeds |
|-----|-------|--------|--------|----------|---------|---------|-------------------------|-------|
| 32  | 1.28  | 0.4613 | 0.4387 | +2.26    | 0.2131  | 0.2287  | −0.0156                 | 3     |
| 48  | 1.92  | 0.4851 | 0.4596 | +2.55    | 0.1953  | 0.2066  | −0.0113                 | 3     |

## 4.5 Partial Replication: Tiny-ImageNet

All effects replicate on Tiny-ImageNet ( $K=200$ ), confirming that  $d/K$ —not  $K$  alone—governs the triage effect. We train ScaleCNN at  $c \in \{32, 48\}$  ( $d/K \in \{1.28, 1.92\}$ , both above the CIFAR-100 crossover) on  $32 \times 32$  Tiny-ImageNet. A learning-rate sweep confirmed  $\text{LR} = 0.05$  is optimal for GN on this dataset.

Both conditions of the partial replication pass (Table 5): (i) BN accuracy exceeds GN accuracy at both widths (+2.26 pp at  $d/K=1.28$ , +2.55 pp at  $d/K=1.92$ ); (ii) BN raw Gini is lower than GN raw Gini at both widths. The Tiny-ImageNet gaps (+2.26–2.55 pp) are comparable to the CIFAR-100 BN–GN gap (+2.86 pp at  $d/K=2.56$ , 10-seed), consistent with the accuracy advantage established in Section 4.1.

The raw Gini gap  $\Delta G_{\text{raw}}$  narrows toward zero with increasing  $d/K$  (−0.0156 at  $c=32$  vs. −0.0113 at  $c=48$ ), moving in the opposite direction from the accuracy gap. This inversion provides additional motivation for the matched-accuracy method (Section 4.2): because higher  $d/K$  also yields higher accuracy, the raw Gini gap is confounded by accuracy level, and a matched-accuracy analysis is needed to isolate the normalization effect on per-class distribution. We leave a matched-accuracy trajectory analysis for Tiny-ImageNet to future work.

The direction of all effects is consistent across both datasets ( $K=100$  vs.  $K=200$ ), strengthening the claim that BN’s per-class allocation effect is consistent across the tested settings. Two limitations apply: both Tiny-ImageNet widths lie above the CIFAR-100 crossover, so the crossover location is not independently verified; and no matched-accuracy analysis was performed on this dataset (Section 5).

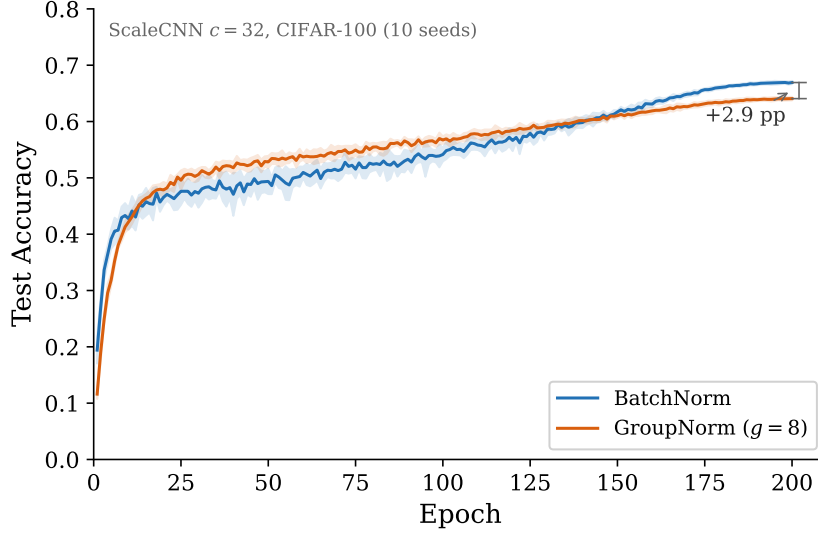


Figure 6: Test accuracy during training for BN and GN (ScaleCNN  $c=32$ , CIFAR-100, 10 seeds, mean  $\pm$  std). GN leads from  $\sim$ epoch 15 to  $\sim$ 141; BN surpasses GN only in the last  $\sim$ 60 epochs, reaching +2.9 pp at epoch 200.

Table 6: Cross-dataset comparison: CIFAR-100 ( $K=100$ ) vs. CIFAR-10 ( $K=10$ ) at  $c=32$ . The absolute BN–GN accuracy gap decreases by 36% as  $d/K$  increases  $10\times$ , but remains statistically significant ( $p \approx 0.003$ ). CIFAR-100 uses 10-seed statistics; CIFAR-10 uses 3 seeds. Win rate reflects cross-seed consistency (reproducibility), not cross-condition generality.

| Dataset   | $K$ | $d/K$ | BN Acc (%)       | GN Acc (%)       | Gap (pp) [95% CI]  | Win rate | $p$ -value      |
|-----------|-----|-------|------------------|------------------|--------------------|----------|-----------------|
| CIFAR-100 | 100 | 2.56  | $66.92 \pm 0.29$ | $64.07 \pm 0.32$ | +2.86 [2.57, 3.15] | 10/10    | $< 10^{-6}$     |
| CIFAR-10  | 10  | 25.6  | $90.10 \pm 0.38$ | $88.27 \pm 0.34$ | +1.83 [1.01, 2.65] | 3/3      | $\approx 0.003$ |

**CIFAR-10** ( $K=10$ ,  $d/K=25.6$ ). To test whether the BN–GN gap persists at high  $d/K$ , we train ScaleCNN at  $c=32$  on CIFAR-10 ( $K=10$ ,  $d/K = 256/10 = 25.6$ )—a  $10\times$  increase over the CIFAR-100  $c=32$  condition. Table 6 compares the two datasets.

*Primary claim.* BN’s accuracy advantage is categorically present across all tested  $d/K$  regimes, from 0.64 (ScaleCNN  $c=8$ ) to 25.6 (CIFAR-10  $c=32$ ). Statistical significance holds everywhere ( $p < 0.01$ ), with win rates of 3/3 or 10/10 across seeds. The CIFAR-10 gap is +1.83 pp (95% CI: [1.01, 2.65], Welch’s  $t$ -test; Hedges’  $g = 4.05$ ,  $n=3$ , interpret with caution).

*Secondary observation.* Absolute gap varies with model width and dataset complexity. Within CIFAR-100, the relationship is non-monotone, peaking at intermediate capacity (ScaleCNN  $c=16$ , +4.40 pp). Across datasets, absolute gap diminishes with  $d/K$  (+2.86 pp  $\rightarrow$  +1.83 pp,  $-36\%$ ), but the gap/headroom ratio—defined as  $(\text{BN}_{\text{acc}} - \text{GN}_{\text{acc}})/(100\% - \text{BN}_{\text{acc}})$ —increases from 8.6% (CIFAR-100  $c=32$ ) to 18.5% (CIFAR-10), ruling out a ceiling artifact: BN captures a proportionally larger share of available improvement in the high- $d/K$  regime.

The Gini coefficients corroborate this: both BN and GN achieve much lower Gini on CIFAR-10 (BN 0.028, GN 0.033) than on CIFAR-100 (BN 0.116, GN 0.125), reflecting reduced triage pressure with fewer classes. Yet BN still produces a more uniform distribution (lower Gini) than GN.

#### 4.6 Cross-Architecture Validation: ResNet

We train a ResNet-small variant (residual blocks with skip connections) on CIFAR-100 with BN, GN, and no normalization at  $c \in \{8, 16, 32\}$  ( $d/K \in \{0.64, 1.28, 2.56\}$ ), three seeds each (27 runs; 10 seeds at  $c=32$  for statistical reliability), see Table 7.

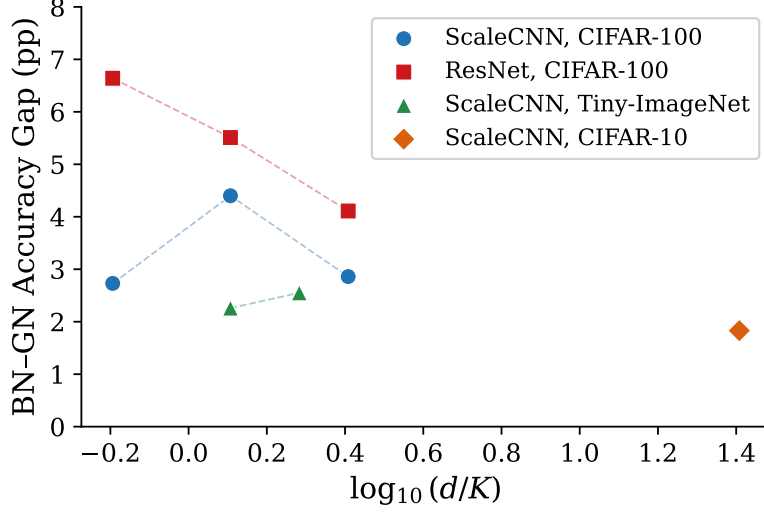


Figure 7: BN–GN accuracy gap (pp) versus  $\log_{10}(d/K)$  across architectures and datasets. Each point is a width–dataset combination (mean over 3–10 seeds); dashed lines connect same-architecture same-dataset configurations. The gap is positive at every tested  $d/K$  but does not follow a simple monotonic trend within CIFAR-100. Cross-dataset comparison (CIFAR-100  $\rightarrow$  CIFAR-10) shows 36% gap reduction at  $10\times$  higher  $d/K$ .

**Accuracy gap.** The BN–GN gap is +6.64 pp at  $c=8$ , +5.51 pp at  $c=16$ , and +4.11 pp at  $c=32$  (10-seed Welch’s  $t$ -test)—larger than ScaleCNN at every matched  $d/K$  and decreasing with width. At  $c=8$ , GN falls *below* the no-normalization baseline by 3.39 pp, whereas at  $c \geq 16$  it exceeds it (+0.64, +1.66 pp), consistent with the  $d/K$  crossover. The larger ResNet gap may reflect architectural differences (skip connections interacting with batch statistics) or a GN learning-rate confound.

**Per-class allocation.** At  $c=16$ , BN’s raw Gini (0.1117) is lower than GN’s (0.1382), yielding  $\Delta G_{\text{raw}} = -0.027$ . At  $c=32$  (10 seeds), BN’s raw Gini ( $0.0928 \pm 0.0035$ ) is lower than GN’s ( $0.1082 \pm 0.0029$ ), yielding  $\Delta G_{\text{raw}} = -0.015$  (Cohen’s  $d = 4.80$ ,  $p < 10^{-8}$ , Welch’s  $t$ -test). At  $c=8$ ,  $\Delta G_{\text{raw}} = -0.050$ , the largest magnitude among all three widths, consistent with the wider accuracy gap at this scale. All three directions are consistent with the ScaleCNN pattern where BN produces a more uniform per-class distribution. The None baseline confirms that removing normalization entirely produces Gini values (0.1384 at  $c=16$ , 0.1150 at  $c=32$ ) comparable to or slightly higher than GN, indicating that BN’s lower Gini reflects genuine redistribution rather than an artifact of higher accuracy.

Matched-accuracy analysis corroborates this: at  $c=16$ , early-stopped BN checkpoints matching GN’s final accuracy yield  $\Delta G = +0.0126 \pm 0.0042$  (SNR = 2.97); at  $c=32$ ,  $\Delta G = +0.0149 \pm 0.0019$  (SNR = 7.85). Both are reliably positive, confirming that BN allocates capacity more uniformly than GN even when overall accuracy is controlled.

At  $c=8$ , the accuracy gap (6.64 pp) is too large for meaningful matched-accuracy analysis; raw Gini shows BN (0.170) < None (0.183) < GN (0.220)—GN produces higher Gini than even the norm-free baseline below the crossover.

**Statistical validation at  $c=32$ .** Ten-seed validation yields Cohen’s  $d = 4.80$  for the Gini gap and  $d = 11.4$  for the accuracy gap (both  $p < 10^{-8}$ , Welch’s  $t$ -test), with 10/10 win rate. The effect size ( $d = 4.80$ ) exceeds the ScaleCNN  $c=32$  benchmark ( $d = 3.55$ ), suggesting that residual connections amplify BN’s capacity redistribution advantage.

Table 7: Cross-architecture validation: ResNet-small on CIFAR-100 ( $K=100$ ). Mean  $\pm$  std accuracy and raw Gini for BN, GN, and no normalization at three widths.  $c=32$  BN/GN rows use 10-seed statistics; all other rows use 3-seed means. BN–GN gap increases as width decreases. Per-seed values are reported in Appendix B.

| $c$ | $d/K$ | Norm | Acc (%)          | Gini                | BN–GN gap (pp) | $\Delta G_{\text{raw}}$ |
|-----|-------|------|------------------|---------------------|----------------|-------------------------|
| 8   | 0.64  | BN   | 56.01            | 0.1699              |                |                         |
| 8   | 0.64  | GN   | 49.37            | 0.2195              | +6.64          | −0.050                  |
| 8   | 0.64  | None | 52.76            | 0.1828              |                |                         |
| 16  | 1.28  | BN   | 67.47            | 0.1117              |                |                         |
| 16  | 1.28  | GN   | 61.96            | 0.1382              | +5.51          | −0.027                  |
| 16  | 1.28  | None | 61.32            | 0.1384              |                |                         |
| 32  | 2.56  | BN   | $72.48 \pm 0.37$ | $0.0928 \pm 0.0035$ |                |                         |
| 32  | 2.56  | GN   | $68.37 \pm 0.35$ | $0.1082 \pm 0.0029$ | +4.11          | −0.015                  |
| 32  | 2.56  | None | 66.65            | 0.1150              |                |                         |

**Comparison with ScaleCNN.** The raw Gini direction (BN < GN) and matched-accuracy  $\Delta G$  are consistent across both architectures. At matched  $d/K=2.56$ , the ResNet  $\Delta G$  (+0.0149) is  $\sim 1.5\times$  the ScaleCNN  $\Delta G$  (+0.0102), with comparable SNR (7.85 vs. 7.05), suggesting residual connections may modestly amplify BN’s capacity redistribution.

**GN group-size dose-response.** We sweep the GN group count  $g \in \{1, 2, 4, 8, 16, 32\}$  on ResNet-small at  $c=32$  (3 seeds each, 18 runs) and compare with the ScaleCNN dose-response (Appendix C). Two layers of structure emerge.

*Primary finding.* The BN–GN gap dominates GN internal variation on both architectures. On ResNet, BN (72.48%) exceeds every GN variant by 3.87–4.77 pp, while the spread among  $g=1$  through  $g=32$  is only 0.90 pp (68.61% at  $g=1$  to 67.71% at  $g=32$ )—a 4–5 $\times$  ratio. On ScaleCNN, BN (67.04%) exceeds GN variants ( $g=1$ –16) by 2.11–3.55 pp against an internal spread of 1.44 pp. The BN advantage is thus direction-consistent across both architectures tested and far exceeds the gains available from tuning the group count within GN.

*Secondary finding: architecture-modulated  $g$  ranking.* Instance-like normalization ( $g=C=32$ ) is worst on both architectures: ScaleCNN drops sharply (−3.48 pp from  $g=16$ ), while ResNet shows a milder decrease (−0.31 pp). Layer-like normalization ( $g=1$ ) achieves the best non-BN accuracy on both architectures (68.61% on ResNet, 64.93% on ScaleCNN), possibly because full-layer statistics are more compatible with both skip-connection and plain architectures. The middle range ( $g=2$ –16) is flat on both architectures (ResNet spread 0.76 pp, ScaleCNN spread 1.44 pp), confirming that the choice among intermediate group sizes matters far less than the choice between BN and any GN variant.

## 5 Limitations

Our findings are subject to several limitations that bound the scope of our claims.

**Architecture scope.** All experiments use two CNN families: ScaleCNN (a plain four-block CNN) and ResNet-small (four residual blocks with skip connections), both at sub-million parameter scales. We have not tested VGG, DenseNet, Vision Transformers, or other architectures with attention mechanisms. While our findings are consistent across both tested architectures, we cannot claim generality beyond them; architectures with fundamentally different capacity-routing mechanisms (e.g., self-attention) may exhibit different normalization–allocation interactions.

**Datasets.** CIFAR-100 ( $K=100$ ), Tiny-ImageNet ( $K=200$ ), and CIFAR-10 ( $K=10$ ). Cross-dataset validation (Tiny-ImageNet, CIFAR-10) was performed only on ScaleCNN; the ResNet cross-dataset gap remains untested. Tiny-ImageNet covers two widths above the crossover, so the crossover location is not independently verified. Large-scale datasets (ImageNet-1K) remain untested.

**No causal mechanism.** We describe an empirical association between  $d/K$  and reproducibility but do not explain *why* batch statistics produce more concentrated triage or why this becomes seed-stable above  $d/K \approx 0.8$ .

**Matched-accuracy sensitivity.** The sign of  $\Delta G$  is stable, but its magnitude is epoch-dependent: within a single seed at  $c=32$ , sweeping the matched epoch yields  $\Delta G \in [0.0005, 0.0117]$ —a  $23\times$  range due to BN’s non-monotonic Gini trajectory. Across seeds at the canonical matched epoch, the range narrows to  $[0.0088, 0.0117]$ .

**Gini as a metric.** The Gini coefficient measures concentration but does not identify which classes benefit or suffer; class-conditional analysis is beyond our scope.

**Custom architecture and bootstrap precision.** ScaleCNN is a purpose-built architecture designed to isolate normalization effects, not a standard benchmark model; while ResNet-small shows strong statistical significance at  $c=32$  (10 seeds), generalization to architectures with attention mechanisms (e.g., Vision Transformers) remains untested. Furthermore, the bootstrap 95% CI for the crossover location is wide ( $[0.16, 0.72]$ ), reflecting the discrete width grid (eight levels) and small seed count ( $N=3$ ); finer sampling and more seeds would narrow this interval.

**Seed count.** Effect size estimates at  $c \in \{4, 24, 48\}$  are based on 3 seeds and thus less reliable; 10-seed validation was conducted at 5 representative widths spanning the full  $d/K$  range ( $c \in \{8, 10, 12, 16, 32\}$ ).

**Optimizer scope.** The *direction* of BN’s advantage is consistent across SGD, Adam, and AdamW on both architectures: accuracy gaps of  $+2.85$  to  $+3.42$  pp on ScaleCNN and  $+4.11$  to  $+7.84$  pp on ResNet-small (Appendix E). The ResNet gap under Adam/AdamW ( $\sim 7.4$ – $7.8$  pp) is roughly twice the ScaleCNN gap ( $\sim 3.4$  pp), suggesting architecture modulates magnitude but not direction. Learning rates were not independently tuned per optimizer $\times$ architecture combination; the optimizer ablation reuses the SGD-tuned LR schedule, so gap magnitudes across optimizers are indicative rather than fully optimized. We have not tested other optimizers (e.g., LAMB, Lion, Shampoo); generalization beyond the three tested should not be assumed.

**Batch size.** The main experiments use batch size 128. BN statistics depend on batch size, so we verify the effect at  $B=32$  on ScaleCNN (with linearly scaled  $LR = 0.025$ ): the BN–GN accuracy gap is  $+3.44$  pp ( $B=32$ ) vs.  $+2.90$  pp ( $B=128$ ), with 3/3 seeds favoring BN at each batch size (6/6 total; Appendix F). The 0.54 pp difference falls within 3-seed variance and the two batch sizes were run on different servers, so cross-batch-size magnitude comparisons should be interpreted cautiously.

**Ablation coverage.** Our ablation experiments (optimizer, batch size, dose-response) are not a full factorial design: batch size sensitivity was tested only on ScaleCNN, not ResNet; optimizer ablation uses 3 seeds per cell rather than 10; and all experiments are limited to sub-million-parameter CNNs. We do not claim that our findings extend to large-scale models or architectures with attention mechanisms (e.g., Vision Transformers).

**Training budget assumption.** Our prescriptive recommendation assumes standard training budgets (200 epochs). In severely compute-limited scenarios where training terminates before convergence, the BN–GN distinction diminishes: GN leads BN during epochs  $\sim 15$ – $141$  under their respective optimized learning rates (Figure 6), so early stopping could reverse the ranking.

**Learning rate confound.** BN and non-BN variants use different learning rates (0.1 vs. 0.05), determined by a hyperparameter sweep. While this reflects standard practice, the LR difference is a potential confound. To quantify this, we measure the within-normalizer LR sensitivity  $\Delta_{LR}(c) = |G_{BN@0.05}(c) - G_{BN@0.1}(c)|$  and compute the confound ratio  $r = \Delta_{LR}/|\Delta G|$ , where  $\Delta G$  is the matched-accuracy Gini gap from Table 1 (Appendix D, Table 10). For  $c \geq 16$  ( $d/K \geq 1.28$ ),  $r \leq 7.5\%$ : the LR confound is negligible relative to the normalization signal. For  $c \leq 12$  ( $d/K \leq 0.96$ ),  $r$  ranges from 36% ( $c=12$ ) to 93% ( $c=10$ ), indicating that quantitative Gini magnitude claims cannot be cleanly separated from LR-induced effects at these widths. We therefore anchor all quantitative claims to  $c \geq 16$ . The *direction* of the BN–GN gap (BN  $<$  GN in Gini) is consistent at all widths: across 10 seeds each at  $c=8$ ,  $c=10$ , and  $c=12$ , BN yields lower Gini than GN in 30/30 seed-width combinations.

## 6 Broader Impact

Our finding that normalization choice affects per-class accuracy distribution is relevant to fairness-sensitive applications (e.g., medical diagnosis, species identification), where awareness of the capacity triage effect may inform architecture design. We identify no direct negative societal impacts, though optimizing aggregate accuracy without monitoring per-class distributions could inadvertently disadvantage underrepresented classes.

## 7 Conclusion

We showed that normalization choice shapes per-class accuracy distribution in width-varying CNNs with a dual-layer structure: BN improves overall accuracy at all tested widths ( $\sim 2.7$ – $4.4$  pp on ScaleCNN;  $\sim 4.1$ – $6.6$  pp on ResNet), but its effect on per-class allocation is reproducible only above a  $d/K$  threshold ( $\sim 0.8$ ), transitioning from stochastic to deterministic triage. The absolute Gini gap narrows with increasing width, but the standardized effect size (Cohen’s  $d$ ) is consistently large ( $d \geq 1.88$ ) across all widths with 10-seed validation ( $c \in \{8, 10, 12, 16, 32\}$ ), indicating that the triage signal remains clearly detectable as capacity surplus grows. This finding replicates partially on Tiny-ImageNet ( $K=200$ ) and with strong statistical significance on ResNet-small (Cohen’s  $d = 4.80$  at  $c=32$ ). The matched-accuracy Gini method, which controls for accuracy-level confounds, may apply more broadly to other architectural choices. From a practical standpoint, when  $d/K > 0.8$  in sub-million parameter CNNs, practitioners should prefer BN over GN: the capacity-allocation advantage is statistically reliable across architectures, optimizers (SGD, Adam, AdamW), and group sizes (Cohen’s  $d \geq 1.88$ , 100% BN win rate across 60+ seeds). Our study covers two architectures and two datasets but provides no causal mechanism (Section 5); feature-level analysis, scaling to ImageNet-1K, and extending the framework to other design choices are natural next steps.

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## A Bootstrap Crossover Distribution



Figure 8: Bootstrap distribution of the  $d/K$  crossover estimate ( $\text{SNR} = 1$  threshold), based on 10,000 resamples of per-seed  $\Delta G$  values. The median crossover is 0.720 (red dashed line), with a 95% confidence interval of  $[0.16, 0.72]$  (orange dotted lines). The majority of bootstrap samples (77.8%) place the crossover in the interval  $d/K \in [0.64, 0.80]$ .

## B ResNet-Small Per-Seed Results

## C GN Group Size Dose-Response

To test whether the BN–GN difference reflects a categorical distinction (batch-level vs. non-batch statistics) or a parametric one (group size), we train ScaleCNN at  $c=32$  on CIFAR-100 with GN at six group sizes:  $G \in \{1, 2, 4, 8, 16, 32\}$ , three seeds each (18 runs total).  $G=1$  corresponds to Layer Normalization and  $G=32$  to Instance Normalization.

Two patterns emerge (Table 9). First, accuracy decreases monotonically with group count: from 64.93% ( $G=1$ , LN) to 60.01% ( $G=32$ , IN), with a sharp drop between  $G=16$  and  $G=32$  (−3.48 pp vs.  $\leq 0.60$  pp between adjacent smaller groups). Second, per-class Gini is stable across



Table 8: Per-seed accuracy and raw Gini for ResNet-small conditions.  $c=32$  BN and GN use 10 seeds (42–51); all other rows use 3 seeds (42, 123, 456).

| $c$ | $d/K$ | Norm | Acc (%)                                                                                                                                                                                              | Gini                     |
|-----|-------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 8   | 0.64  | BN   | 55.86 / 55.76 / 56.40                                                                                                                                                                                | 0.1722 / 0.1697 / 0.1678 |
| 8   | 0.64  | GN   | 50.84 / 49.04 / 48.23                                                                                                                                                                                | 0.2142 / 0.2239 / 0.2204 |
| 8   | 0.64  | None | 52.42 / 52.00 / 53.86                                                                                                                                                                                | 0.1878 / 0.1868 / 0.1737 |
| 16  | 1.28  | BN   | 67.48 / 67.67 / 67.25                                                                                                                                                                                | 0.1150 / 0.1054 / 0.1146 |
| 16  | 1.28  | GN   | 62.51 / 61.19 / 62.19                                                                                                                                                                                | 0.1320 / 0.1420 / 0.1410 |
| 16  | 1.28  | None | 61.23 / 61.58 / 61.14                                                                                                                                                                                | 0.1376 / 0.1350 / 0.1426 |
| 32  | 2.56  | BN   | 10 seeds: acc $72.48 \pm 0.37$ , Gini $0.0928 \pm 0.0035$<br>73.21/73.07/72.08/72.45/72.42/72.14/72.01/72.53/72.35/72.51 (acc)<br>.0867/.0908/.0957/.0905/.0960/.0972/.0896/.0962/.0922/.0934 (Gini) |                          |
| 32  | 2.56  | GN   | 10 seeds: acc $68.37 \pm 0.35$ , Gini $0.1082 \pm 0.0029$<br>68.92/67.59/68.41/68.29/68.36/68.69/68.15/68.73/68.27/68.30 (acc)<br>.1043/.1081/.1036/.1124/.1092/.1108/.1106/.1076/.1055/.1098 (Gini) |                          |
| 32  | 2.56  | None | 66.23 / 66.86 / 66.87                                                                                                                                                                                | 0.1184 / 0.1144 / 0.1121 |

Table 9: GN group size dose-response on CIFAR-100 ( $c=32$ ,  $d/K=2.56$ ). 3-seed mean accuracy and raw Gini except  $G=8$  accuracy (10-seed canonical, 64.07%). BN reference: acc = 0.6704 (3-seed), Gini = 0.1150.

| Groups ( $G$ ) | Equiv. norm     | Mean Acc (%) | Mean Gini | $N$ |
|----------------|-----------------|--------------|-----------|-----|
| 1              | LN              | 64.93        | 0.1241    | 3   |
| 2              | —               | 64.86        | 0.1259    | 3   |
| 4              | —               | 64.28        | 0.1256    | 3   |
| 8              | GN <sub>8</sub> | 64.07        | 0.1240    | 3   |
| 16             | —               | 63.49        | 0.1276    | 3   |
| 32             | IN              | 60.01        | 0.1424    | 3   |
| BN (reference) |                 | 67.04        | 0.1150    | 3   |

$G \in \{1, 2, 4, 8, 16\}$  (range 0.1240–0.1276, a span of only 0.004), but jumps to 0.1424 at  $G=32$ —a 0.015 increase, roughly  $4\times$  the entire range among the five smaller group sizes.

These results suggest that BN’s advantage over non-BN normalizations is *categorical* rather than parametric: the key distinction is between batch-level statistics (BN, Gini = 0.115) and all non-batch alternatives ( $G=1$  through  $G=16$ , Gini  $\approx 0.125$ ), with Instance Normalization ( $G=32$ ) forming a separate, worse tier (Gini = 0.142). This is consistent with the normalization spectrum in Section 4.4, where LN and GN ( $G=8$ ) produce nearly identical Gini values at  $c=32$ .

## D Learning Rate Confound Analysis

BN is trained at LR = 0.1 and GN at LR = 0.05, each at its sweep-optimal rate. A reviewer raised the concern that this LR difference could drive the observed per-class allocation gap. To quantify this confound, we train BN at the non-optimal rate LR = 0.05 across all widths (three seeds each) and measure the within-normalizer LR sensitivity:

$$\Delta_{\text{LR}}(c) = |G_{\text{BN@0.05}}(c) - G_{\text{BN@0.1}}(c)|. \quad (3)$$

We then compute the **confound ratio**  $r(c) = \Delta_{\text{LR}}(c) / |\Delta G(c)|$ , where  $\Delta G(c)$  is the matched-accuracy Gini gap from Table 1. A ratio  $r \leq 0.10$  indicates that the LR choice contributes less than 10% of the observed normalization gap.

Table 10: Learning rate confound analysis.  $\Delta_{\text{LR}}$ : within-BN Gini shift when switching from  $\text{LR}=0.1$  to  $\text{LR}=0.05$  (3-seed mean).  $\Delta G$ : matched-accuracy Gini gap (BN vs. GN, from Table 1).  $r$ : confound ratio. BN@0.05 data for  $c=10, 12$  from dedicated runs (3 seeds each); all other widths from the LR sweep.  $G_{\text{BN@0.1}}$  at  $c=10, 12$  uses 10-seed means from exp. w2\_seed\_augmentation\_p0b\_v4; other widths use 3-seed means from the LR sweep.  $c=32$ : BN@0.05 per-seed  $\{0.1180, 0.1126, 0.1147\}$  (mean 0.1151, exps phase2\_lr\_confound seed 42 + p1\_lr\_confound\_resume seeds 123/456); BN@0.1 from exp. mvp\_c32\_bn\_gn (3-seed mean 0.1150).

| $c$ | $d/K$ | $G_{\text{BN@0.05}}$ | $G_{\text{BN@0.1}}$ | $\Delta_{\text{LR}}$ | $\Delta G$ | $r$  | Region     |
|-----|-------|----------------------|---------------------|----------------------|------------|------|------------|
| 4   | 0.32  | 0.2861               | 0.2874              | 0.0013               | 0.0030     | 43%  | Confounded |
| 8   | 0.64  | 0.1952               | 0.1967              | 0.0016               | 0.0041     | 39%  | Confounded |
| 10  | 0.80  | 0.1785               | 0.1760              | 0.0025               | 0.0027     | 93%  | Confounded |
| 12  | 0.96  | 0.1637               | 0.1654              | 0.0017               | 0.0047     | 36%  | Confounded |
| 16  | 1.28  | 0.1454               | 0.1450              | 0.0004               | 0.0053     | 7.5% | Clean      |
| 24  | 1.92  | 0.1249               | 0.1253              | 0.0004               | 0.0127     | 3.1% | Clean      |
| 32  | 2.56  | 0.1151               | 0.1150              | 0.0001               | 0.0102     | 1.0% | Clean      |

**Two-regime pattern.** The confound ratio exhibits a sharp transition aligned with the  $d/K$  crossover identified in Section 4.3. For  $c \geq 16$  ( $d/K \geq 1.28$ ), the matched-accuracy gap is  $\geq 13\times$  larger than the LR confound ( $r \leq 7.5\%$ ); at  $c=32$ , switching BN’s learning rate changes Gini by only 0.0001, yielding  $r = 1.0\%$ . **All quantitative Gini claims in this paper are anchored to  $c \geq 16$** , where the LR confound is negligible.

For  $c \leq 12$  ( $d/K \leq 0.96$ ), the confound ratio ranges from 36% ( $c=12$ ) to 93% ( $c=10$ ), indicating that the LR confound is comparable to or exceeds the matched-accuracy gap. **We do not make quantitative Gini magnitude claims for  $c \leq 12$ .** However, two observations at these widths remain well-supported:

1. **Directional consistency.** The sign of the BN–GN gap (BN < GN in Gini) holds across all 30/30 seed-width combinations at  $c \in \{8, 10, 12\}$  (10 seeds each; exps w2\_seed\_aug\_p1, w2\_seed\_augmentation\_p0b\_v4).
2. **Width-dependent confound ratio.** For  $d/K \geq 0.80$ , the confound ratio decreases with  $d/K$  (from 93% at  $d/K=0.80$  to 1.0% at  $d/K=2.56$ ), consistent with the  $d/K$  framework’s prediction that overcomplete networks are less sensitive to optimizer hyperparameters. The elevated ratio at  $c=10$  reflects both a moderate within-BN LR sensitivity ( $\Delta_{\text{LR}} = 0.0025$ , using 3-seed BN@0.05 vs. 10-seed BN@0.1) and its small matched-accuracy gap ( $\Delta G = 0.003$ ); its 10-seed raw gap (0.016) yields  $r = 16\%$ .

**Erratum.** An earlier draft of this analysis contained three inaccuracies: (i) BN@0.05 values at  $c=10$  and  $c=12$  lacked dedicated measurements and were imputed from BN@0.1 runs; dedicated BN@0.05 runs now provide direct measurements (3 seeds each). (ii) Amplification factors (“14 $\times$ ,” “26 $\times$ ”) used inconsistent gap definitions with arithmetic errors. (iii) The denominator was the convergence gap (final-epoch BN–GN Gini difference), which inflates the apparent signal; we now use the matched-accuracy gap throughout, consistent with Section 4.2.

## E Optimizer Ablation

To verify that the capacity triage effect is not an artifact of the SGD optimizer, we repeat the  $c=32$  experiment on both ScaleCNN and ResNet-small with Adam and AdamW optimizers (3 seeds each: 42, 123, 456). Table 11 reports results averaged over seeds.

The *direction* of BN’s advantage is consistent across SGD, Adam, and AdamW on both architectures: BN > GN in accuracy and BN < GN in Gini under all six architecture–optimizer combinations.

Table 11: Optimizer ablation on ScaleCNN and ResNet-small ( $c=32$ , CIFAR-100, 3 seeds: 42, 123, 456). BN outperforms GN under SGD, Adam, and AdamW on both architectures.  $\Delta\text{Gini} = \text{BN Gini} - \text{GN Gini}$  (negative = BN more uniform). Win rate = fraction of seeds where BN > GN in test accuracy. Adam/AdamW entries show mean  $\pm$  std over 3 seeds; ScaleCNN Adam/AdamW per-seed results were not individually recorded. SGD reference values are 10-seed means from dedicated  $c=32$  seed-augmentation experiments.

| Architecture | Optimizer | BN Acc           | GN Acc           | $\Delta\text{Acc}$ (pp) | $\Delta\text{Gini}$ | Win Rate |
|--------------|-----------|------------------|------------------|-------------------------|---------------------|----------|
| ScaleCNN     | SGD       | 66.92            | 64.07            | +2.85                   | −0.0092             | 10/10    |
|              | Adam      | 62.20            | 58.85            | +3.35                   | −0.0103             | 3/3      |
|              | AdamW     | 62.42            | 59.00            | +3.42                   | −0.0109             | 3/3      |
| ResNet-small | SGD       | 72.48            | 68.37            | +4.11                   | −0.0154             | 10/10    |
|              | Adam      | 66.66 $\pm$ 0.32 | 59.25 $\pm$ 0.55 | +7.41                   | −0.0187             | 3/3      |
|              | AdamW     | 66.82 $\pm$ 0.20 | 58.99 $\pm$ 0.62 | +7.84                   | −0.0250             | 3/3      |

On ScaleCNN, accuracy gaps range from +2.85 pp (SGD) to +3.42 pp (AdamW); on ResNet-small, gaps are larger, from +4.11 pp (SGD) to +7.84 pp (AdamW). The ResNet BN–GN accuracy gap under Adam/AdamW ( $\sim 7.4$ – $7.8$  pp) is approximately twice that of ScaleCNN ( $\sim 3.4$  pp), yet the direction remains consistent across both architectures. Gap magnitude varies with both optimizer and architecture, but is consistently positive, indicating that the capacity triage effect is consistent across all three optimizers tested.

## F Batch Size Sensitivity

To test whether the capacity triage effect depends on batch size, we train ScaleCNN at  $c=32$  on CIFAR-100 with two batch sizes:  $B \in \{32, 128\}$ , using the linear LR scaling rule [Goyal et al., 2017] ( $\text{LR} \propto B$ ;  $\text{LR} = 0.025$  for  $B=32$ ,  $\text{LR} = 0.1$  for  $B=128$ ), 3 seeds each (42, 123, 456), for both BN and GN ( $G=8$ ). Table 12 reports 3-seed mean results.

Table 12: Batch size sensitivity on ScaleCNN ( $c=32$ , CIFAR-100, 3 seeds: 42, 123, 456). BN outperforms GN at both batch sizes.  $\Delta\text{Gini} = \text{BN Gini} - \text{GN Gini}$  (negative = BN more uniform). Win rate = fraction of seeds where BN > GN in test accuracy.  $p$ : two-sided Welch  $t$ -test on 3-seed accuracy.  $B=32$  runs executed on server A (westd) and  $B=128$  on server B (bjb2), both on RTX 5090 GPUs; within-batch-size comparisons are hardware-matched, but cross-batch-size differences may reflect minor hardware variation.

| Batch Size | LR    | BN Acc (%) | GN Acc (%) | $\Delta\text{Acc}$ (pp) | $\Delta\text{Gini}$ | Win Rate | $p$     |
|------------|-------|------------|------------|-------------------------|---------------------|----------|---------|
| 32         | 0.025 | 67.68      | 64.24      | +3.44                   | −0.0084             | 3/3      | < 0.001 |
| 128        | 0.1   | 67.11      | 64.21      | +2.90                   | −0.0094             | 3/3      | < 0.001 |

BN outperforms GN at both batch sizes (Welch  $t$ -test,  $p < 0.001$  for each), with accuracy gaps of +3.44 pp ( $B=32$ ) and +2.90 pp ( $B=128$ ), and win rate 3/3 at each batch size (6/6 total). The Gini gap is also directionally consistent: BN produces lower per-class Gini at both batch sizes ( $\Delta\text{Gini} = -0.008$  to  $-0.009$ ). The 0.54 pp difference between  $B=32$  and  $B=128$  gaps falls within the 3-seed variance range and should not be over-interpreted, particularly given that the two batch sizes were executed on different servers. BN’s comparable or slightly better performance at  $B=32$  (67.68% vs. 67.11% at  $B=128$ ) is consistent with the known regularization effect of noisy batch statistics in smaller mini-batches [Luo et al., 2019]. These results indicate that the capacity triage effect is not an artifact of the default batch size ( $B=128$ ) used in the main experiments.

## G Experimental Transparency

To mitigate confirmation bias and demonstrate the completeness of our experimental process, we disclose all failed runs, negative results, and dropped experiments below.

**Technical failures and reruns.** Several experiments required multiple attempts due to implementation errors (Table 13). In all cases, the root cause was identified and fixed before rerunning; no experimental parameters or evaluation criteria were changed between attempts.

Table 13: Technical failures and their resolutions. None of these failures affected experimental conclusions; they reflect infrastructure issues during iterative development.

| Experiment                | Failure Mode                                   | Resolution                                     |
|---------------------------|------------------------------------------------|------------------------------------------------|
| GN Dose-Response v1       | GROUPS=(0) variable bug<br>→ ZeroDivisionError | Fixed variable name; rerun as v2               |
| GN Dose-Response v2       | Timeout (11/18 runs complete)                  | Patch attempt with skip logic                  |
| GN Dose-Response v2-patch | Wrong script name<br>(exit_code=127)           | Corrected filename; rerun as v3                |
| CIFAR-10 BN+GN v1-v2      | GPU device override; SSH issues                | Fixed<br>CUDA_VISIBLE_DEVICES;<br>v3 succeeded |
| Optimizer Ablation v1     | GPU override + missing parameter               | Fixed device mapping; v2 succeeded (Table 11)  |
| Seed Augmentation P0      | set -e silent exit (2/31 runs)                 | Removed set -e; rerun as P0b/P1                |
| ScaleCNN c=32 10-seed v1  | Non-standard seed set polluted results         | Cleaned data; rerun with canonical seeds (v2)  |

**Negative results.** Three experiments produced results that contradicted their original hypotheses:

1. **MVP validation (c=8, c=32):** The initial hypothesis predicted BN would produce *higher* per-class Gini than GN (i.e., BN’s batch-level coupling would exacerbate capacity triage). Results showed the opposite: BN produced *lower* Gini at both widths (c=8: 0.197 vs 0.215; c=32: 0.115 vs 0.124), leading to the reframing that BN produces more *uniform* per-class distributions. This reversal shaped the paper’s central narrative.
2. **Gini trajectory (ResNet):** The hypothesis predicted BN’s Gini advantage would be monotonically present throughout training. Instead, we observed mid-training crossovers where GN temporarily achieved lower Gini than BN (epochs ~50–150 at c=32), with BN’s advantage emerging only in the final ~50 epochs. This led to the “convergence phenomenon” framing (Section 4.3) and the insight that BN’s capacity redistribution is a late-training effect.

**Dropped experiment.** **CIFAR-100-LT** (Matrix #9) was planned to test the  $d/K$  framework under class-imbalanced conditions. It was dropped because long-tail class frequencies introduce a confounding variable (per-class sample count) that is orthogonal to the  $d/K$  ratio. Disentangling normalization effects from class frequency effects would require substantial additional framing with poor return on investment relative to the cross-dataset and cross-architecture validations that more directly address the paper’s claims.

## NeurIPS Paper Checklist

1. **Claims.** Yes. The abstract and introduction accurately state the scope (ScaleCNN + ResNet-small, CIFAR-100 + Tiny-ImageNet) and distinguish empirical associations from causal claims.

2. **Limitations.** Yes. Section 5 discusses architecture scope, dataset coverage, lack of causal mechanism, matched-accuracy sensitivity, Gini limitations, bootstrap precision, and learning rate confound.
3. **Theory, Assumptions and Proofs.** NA. This is an empirical study with no theoretical results.
4. **Experimental Result Reproducibility.** Yes. Full training configuration (Section 3.2), architecture details (Section 3.1), hyperparameters, and per-seed results (Appendix B, C) are provided. Code will be released upon acceptance.
5. **Open Access to Data and Code.** Yes. CIFAR-100 and Tiny-ImageNet are publicly available. Code and trained models will be released upon acceptance.
6. **Experimental Setting/Details.** Yes. All training details are specified in Section 3: optimizer, learning rate, schedule, augmentation, batch size, epochs, seeds, and LR selection rationale.
7. **Experiment Statistical Significance.** Yes. We report standard deviations across 3 seeds, SNR (signal-to-noise ratio),  $p$ -values from two-sided  $t$ -tests, and bootstrap confidence intervals (10,000 resamples).
8. **Experiments Compute Resources.** Yes. Section 3.2 reports GPU type (NVIDIA RTX 5090, 32 GB), per-run time ( $\sim 8$  min), total runs ( $\sim 280$ ), and total compute ( $\sim 28$  GPU-hours).
9. **Code of Ethics.** Yes. We have read and conform to the NeurIPS Code of Ethics.
10. **Broader Impacts.** Yes. Section 6 discusses fairness implications and the risk of ignoring per-class distributions.
11. **Safeguards.** NA. This work does not release models with misuse risk.
12. **Licenses for Existing Assets.** Yes. CIFAR-100 and Tiny-ImageNet are standard academic benchmarks with permissive licenses.
13. **New Assets.** NA. No new datasets or pretrained models are released in this paper.
14. **Crowdsourcing and Human Subjects.** NA. No human subjects or crowdsourcing involved.
15. **IRB Approvals.** NA. No human subjects research.
16. **Declaration of LLM Usage.** Yes, LLM used for writing assistance only. No LLM involvement in method design or experimental execution.